

ProMi: An Efficient Prototype-Mixture Baseline for Few-Shot Segmentation with Bounding-Box Annotations

Challenge

Few-Shot Segmentation with Bounding-Box Annotations

- **Support set S :** One or a few training images and their corresponding sets of **bounding-box annotations** for the class of interest
- **Query set Q :** Images with unknown ground truth **binary segmentation masks**

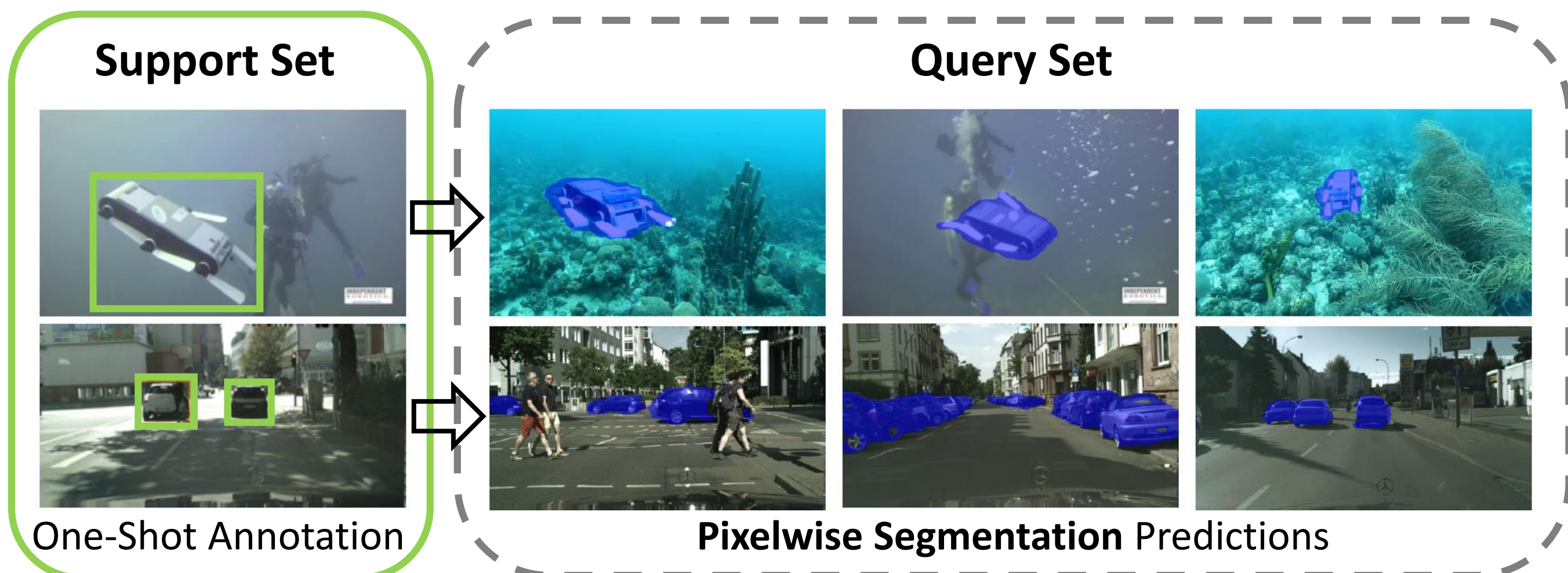
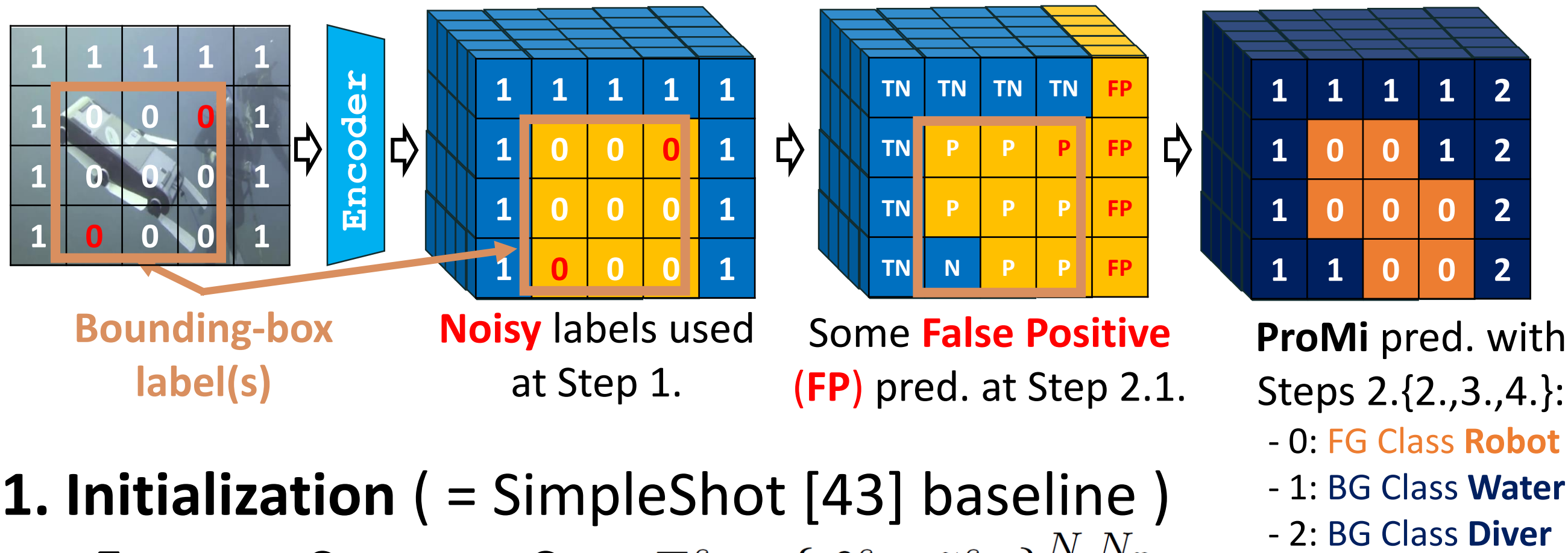


Figure 1. Few-Shot Segmentation with Bounding-Box Annotations using ProMi.

- **Goal:** Segment the query set images at the pixel-level using a few support training data, annotated with bounding-box annotations
- **Proposed label reformulation:** One can consider bounding-box labels as **coarse and noisy pixel-level labels**

Proposed Prototype Mixture Model

Prototype Mixture Estimation



1. Initialization (= SimpleShot [43] baseline)

- Feature Support Set: $F^s = \{f_{im}^s, \tilde{y}_{im}^s\}_{i=1, m=1}^{N, N_p}$
- Trainable Prototypes: $W = \{w_{FG}\} \cup \{w_{BG}^{(k)}\}_{k=1}^K$
- **Prototype initialization:**

$$w_{FG} = \frac{\sum_{i,m} \tilde{y}_{im}^s f_{im}^s}{\sum_{i,m} \tilde{y}_{im}^s} \quad w_{BG}^{(1)} = \frac{\sum_{i,m} (1 - \tilde{y}_{im}^s) f_{im}^s}{\sum_{i,m} (1 - \tilde{y}_{im}^s)}$$

2. Iterative Prototype Refinement (Our Main Contribution)

2.1. Hard-label predictions in the latent space

$$\hat{y}_{im} = \arg \max_k \cos(f_{im}^s, w_k)$$

2.2. Refinement of Background Prototype(s)

$$w_{BG}^{(k)} = \frac{\sum_{i,m} (1 - \tilde{y}_{im}^s) f_{im}^s \mathbb{1}(\hat{y}_{im} = k)}{\sum_{i,m} (1 - \tilde{y}_{im}^s) \mathbb{1}(\hat{y}_{im} = k)}$$

2.3. Adding New Background Prototypes

$$w_{BG}^{(K+1)} = \frac{\sum_{i,m} (1 - \tilde{y}_{im}^s) f_{im}^s \mathbb{1}(\hat{y}_{im} = 0)}{\sum_{i,m} (1 - \tilde{y}_{im}^s) \mathbb{1}(\hat{y}_{im} = 0)}$$

2.4. Refinement of the Foreground Prototype

$$w_{FG} = \frac{\sum_{i,m} \tilde{y}_{im}^s f_{im}^s \mathbb{1}(\hat{y}_{im} = 0)}{\sum_{i,m} \tilde{y}_{im}^s \mathbb{1}(\hat{y}_{im} = 0)}$$

2.5. Stopping Criterion: The iterative process stops when either K_{Max} is reached or no new False Positives are found.

Proposed Framework

A. Training

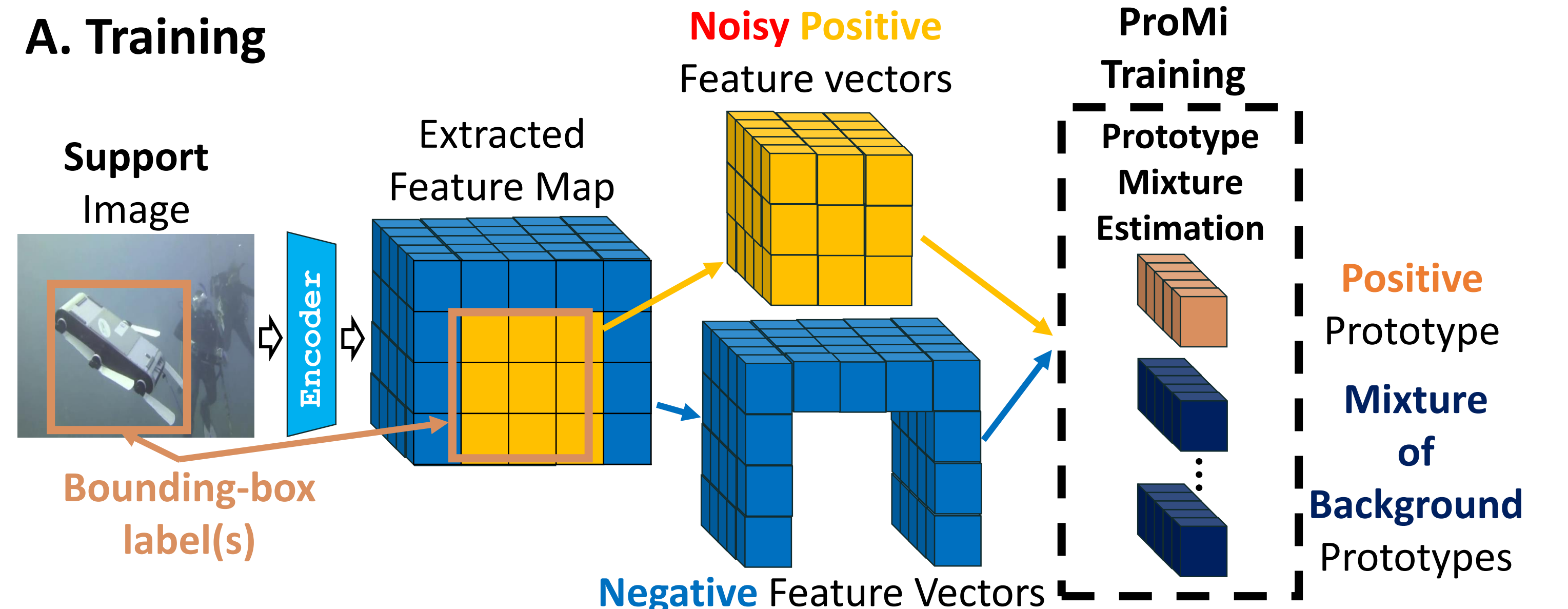


Figure 2. Efficient Training stage.

B. Inference

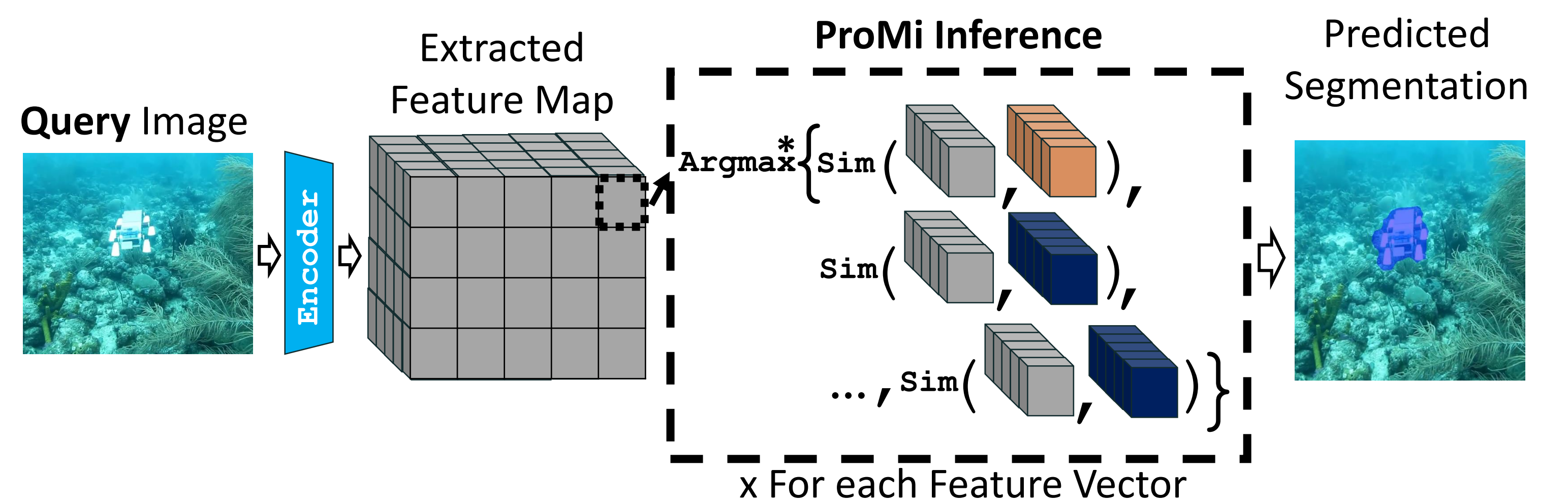


Figure 3. Inference. *Before applying Argmax, we first resize the obtained 2D map of logits using bilinear interpolation.

Encoder: Pre-trained and fixed DNN feature extractor

Sim: Cosine Similarity Metric

Experiments

Quantitative Experiments:

Method	Backbone	PASCAL VOC 2012			MS-COCO		
		1-Shot	5-Shot	10-Shot	1-Shot	5-Shot	10-Shot
BD-CSPN [21]	ResNet-50	43.9	45.9	45.9	23.4	25.5	25.7
SimpleShot [43]	ResNet-50	41.4	44.8	45.5	22.8	25.5	25.7
ProMi (proposed)	ResNet-50	45.4	51.5	55.8	26.1	31.2	33.6
BD-CSPN [21]	DINOv2	39.6	44.3	45.9	18.1	22.4	23.4
SimpleShot [43]	DINOv2	36.7	43.7	45.8	16.5	22.3	23.9
ProMi (proposed)	DINOv2	44.1	50.4	53.2	20.7	28.4	31.2

Table 1. Quantitative results (mean-IoU).

Qualitative Experiments:

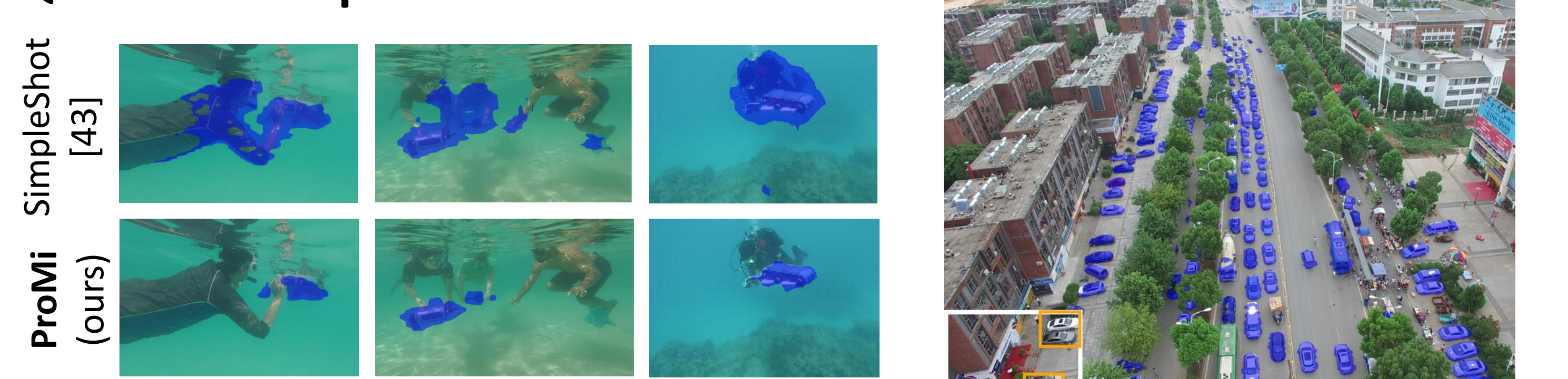


Figure 4. Qualitative comparison for underwater robot segmentation.



Figure 5. One-Shot result on high-resolution UAV data.

Ablation Study on PASCAL VOC with DINOv2 (ViT-B/14) encoder:

BG Mixt.	FG Ref.	1-Shot	5-Shot	10-Shot
×	×	33.4	44.2	46.5
✓	×	40.7	46.0	48.4
✓	✓	44.1	50.4	53.2

Table 2. Ablation of ProMi components BG Mixt. (Steps 2.2 & 2.3) and FG Ref. (Step 2.4) (mean-IoU).

Figure 6. ProMi mean-IoU as a function of K.

Conclusions & Future Work

Strengths:

- Efficient training
- Competitive Performances

Limitations & Ongoing Work:

- Multiclass segmentation
- Prediction resolution refinement

Our code: <https://github.com/ThalesGroup/promi>